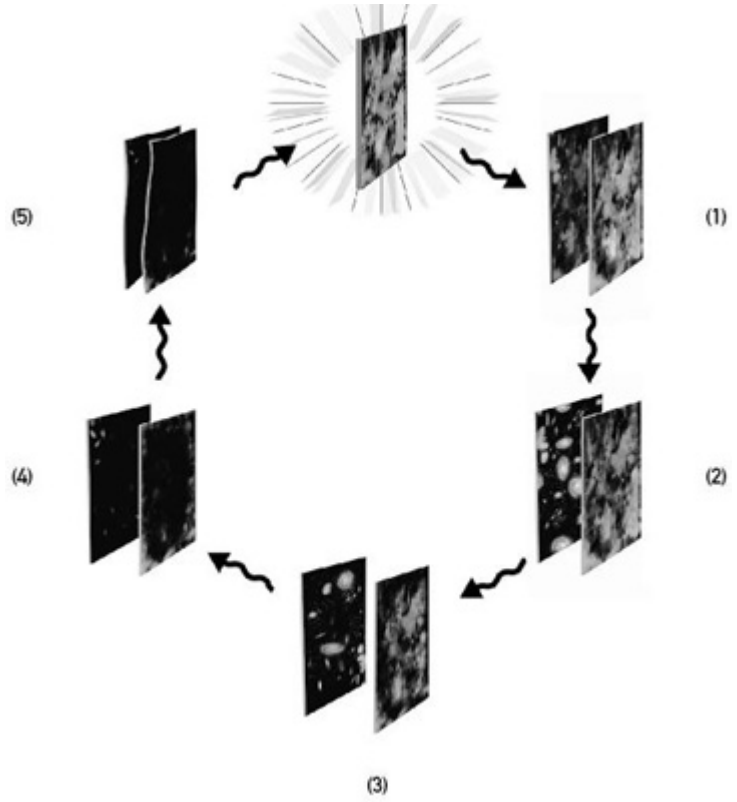


Figure 13.8 Various stages in the cyclic braneworld cosmological model.

By this point, our three-brane has completed its rebound from the initial collision and has started to approach the second three-brane once again. As we get closer and closer to another collision, quantum jitters of the strings attached to our brane overlie its uniform emptiness with tiny ripples, Stage 5. As we continue to pick up speed, the ripples continue to grow; then, in a cataclysmic collision, we smack into the second three-brane, we bounce off, and the cycle starts anew. The quantum ripples imprint tiny inhomogeneities in the radiation and matter produced during the collision and, much as in the inflationary scenario, these deviations from perfect uniformity grow into clumps that ultimately generate stars and galaxies.

These are the major stages in the cyclic model (also known tenderly as the *big splat*). Its premise—colliding braneworlds—is very different from that of the successful inflationary theory, but there are, nevertheless, significant points of contact between the two approaches. That both rely on quantum agitation to generate initial nonuniformities is one essential similarity. In fact, Steinhardt and Turok argue that the equations governing the quantum ripples in the cyclic model are nearly